

Terminal Navigation, Folder Creation & Flags

The command line (Terminal) is a text-based file explorer. Instead of using a mouse to click through windows, you use precise text commands to navigate your file system, build project workspaces, and modify command behaviors using options called flags.

1. Core Navigation & Workspace Creation

Command	Instruction & Core Mechanics	Step-by-Step Code & Output
<code>pwd</code> (Print Working Directory)	Prints out the exact absolute path of your current location. Your digital map coordinates.	<pre>\$ pwd /Users/username/Documents</pre>
<code>ls</code> (List)	Lists the names of all visible files and folders inside your current folder path.	<pre>\$ ls Desktop Documents Downloads</pre>
<code>cd [folder]</code> (Change Directory)	Steps forward into a specified subfolder within your current location.	<pre>\$ cd Documents \$ pwd /Users/username/Documents</pre>
<code>cd ..</code> (Go Back)	Steps backward up out of your current folder into its parent folder. Note the space.	<pre>\$ cd .. \$ pwd /Users/username</pre>
<code>mkdir [name]</code> (Make Directory)	Generates a brand-new, empty folder with your chosen name inside your current path location.	<pre>\$ mkdir python_projects \$ ls Desktop Documents python_projects</pre>

2. Understanding CLI Modifiers & Flags

A command follows a predictable structure: `[Command] [Target] [Flags]`. Flags start with a single dash (`-`) for short letter options or double dashes (`--`) for full-word settings, altering how the command behaves.

Syntax Example	Instruction & Core Mechanics	Step-by-Step Code & Output
<code>ls -l</code> (Long Format Flag)	-l (Long): Modifies <code>ls</code> to output extra structural details like file size, modification dates, and system permissions.	<pre>\$ ls -l -rw-r--r-- 1 user staff 412 Jun 24 script.py drwxr-xr-x 2 user staff 64 Jun 24 data</pre>
<code>ls -a</code> (All Files Flag)	-a (All): Forces <code>ls</code> to display hidden configuration files (items starting with a dot, such as hidden Git configurations).	<pre>\$ ls -agit Documents script.py</pre>
<code>ls -la</code> (Chaining Flags)	Combines multiple single-letter parameters under a single dash to apply multiple behavioral overrides at once.	<pre>\$ ls -la drwxr-xr-x 3 user staff 96 .git -rw-r--r-- 1 user staff 412 script.py</pre>

Syntax Example	Instruction & Core Mechanics	Step-by-Step Code & Output
<code>command -d</code> (Common Debug/Path)	In modern programming tools, a <code>-d</code> flag is routinely used to specify a targeted Directory path or switch an app into Debug mode.	<pre>\$ python script.py -d [DEBUG] Initializing sandbox variables...</pre>